Problem 1

I

B, C, D

II

A, C

III

A, C

IV

A, C

Problem 2

a

Class 1, as the closest point is class 1

b

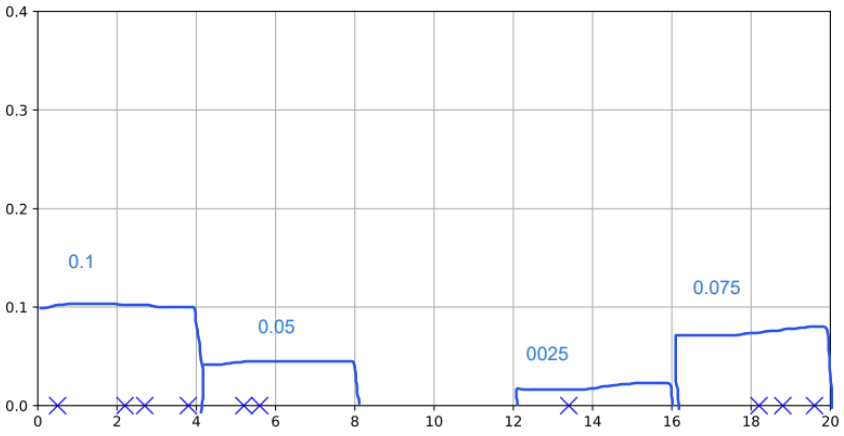
Class 2, as more than half of the 20 closest points are class 2

c

Class 1 because the uniform kernel considers all data points uniformly within d and there are more points of class 1 that are being considered.

Problem 3

a

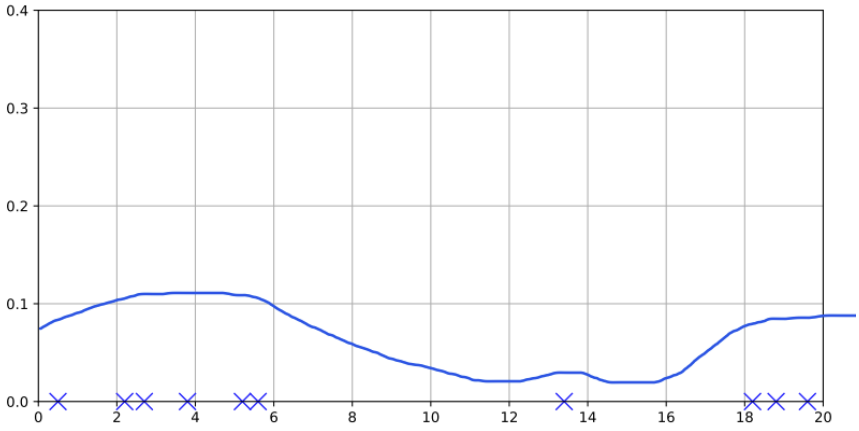


b

0.05, we look at the bucket that $x = 6.0$ would land in, and it has p(x) = 0.05

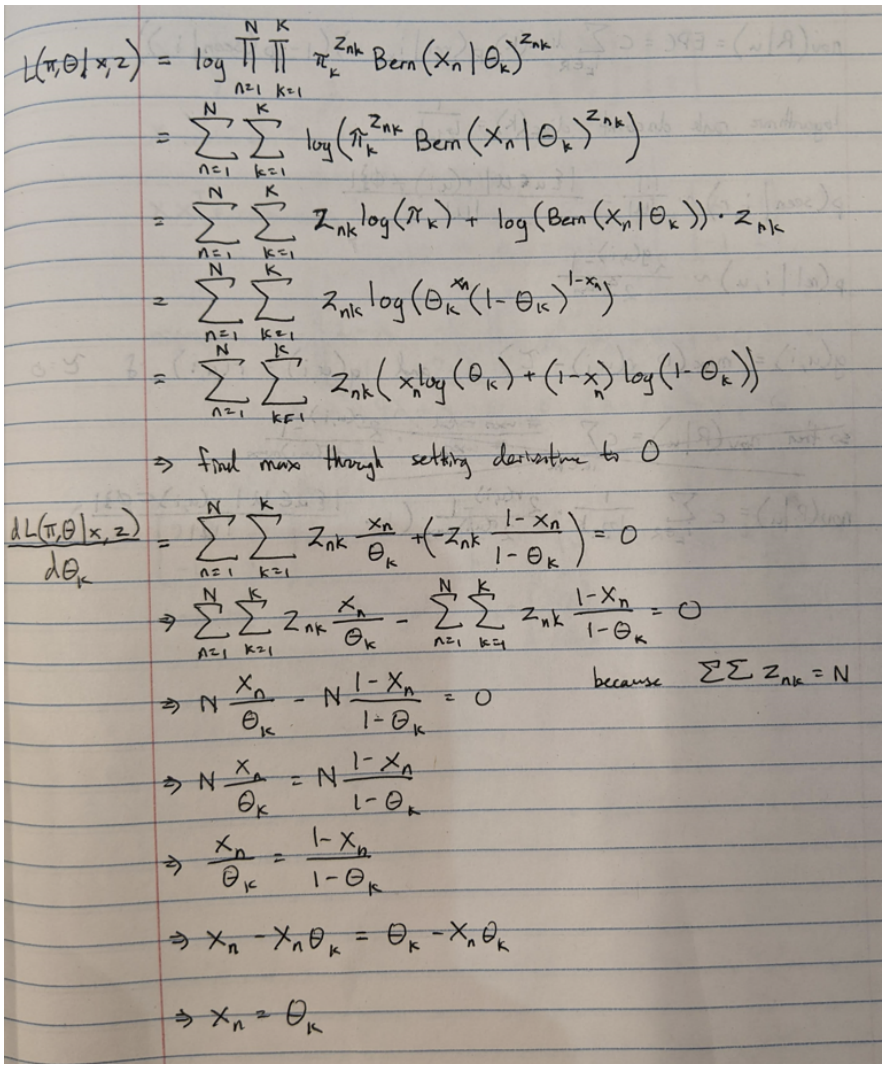
c

Increasing the bucket size generally smoothes out the curve more. Additionally, switching from a naive estimator over to a kernel estimator would also smooth out the curve.

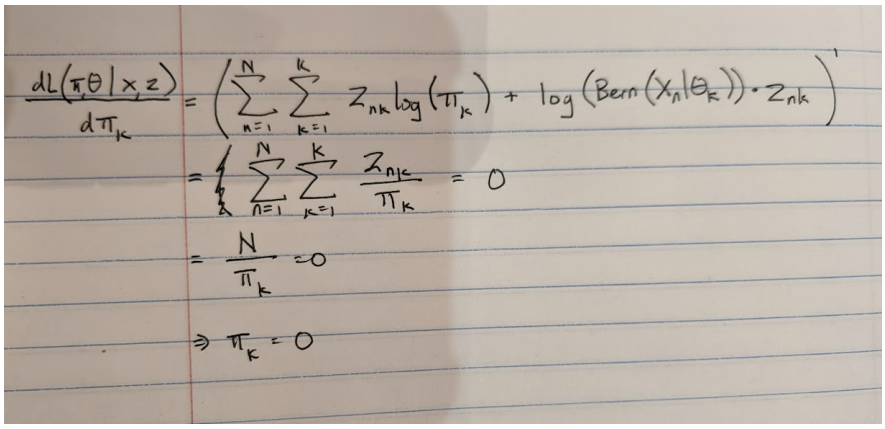


Problem 4

a


$L(\pi,\theta|x,z) = \log \prod_{n=1}^{N}\prod_{k=1}^{K} \pi_k^{z_{nk}} \text{Bern}(x_n|\theta_k)^{z_{nk}}$
$= \sum_{n=1}^{N}\sum_{k=1}^{K} \log\left(\pi_k^{z_{nk}} \text{Bern}(x_n|\theta_k)^{z_{nk}}\right)$
$= \sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\log(\pi_k) + \log(\text{Bern}(x_n|\theta_k)) \cdot z_{nk}$
$= \sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\log(\theta_k^{x_n}(1-\theta_k)^{1-x_n})$
$= \sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\left(x_n\log(\theta_k) + (1-x_n)\log(1-\theta_k)\right)$
⇒ find max through setting derivative to 0
$\frac{dL(\pi,\theta|x,z)}{d\theta_k} = \sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\frac{x_n}{\theta_k} + \left(-z_{nk}\frac{1-x_n}{1-\theta_k}\right) = 0$
$\Rightarrow \sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\frac{x_n}{\theta_k} - \sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\frac{1-x_n}{1-\theta_k} = 0$
$\Rightarrow N\frac{x_n}{\theta_k} - N\frac{1-x_n}{1-\theta_k} = 0$ because $\sum\sum z_{nk} = N$
$\Rightarrow N\frac{x_n}{\theta_k} = N\frac{1-x_n}{1-\theta_k}$
$\Rightarrow \frac{x_n}{\theta_k} = \frac{1-x_n}{1-\theta_k}$
$\Rightarrow x_n - x_n\theta_k = \theta_k - x_n\theta_k$
$\Rightarrow x_n = \theta_k$

b


$\frac{dL(\pi,\theta|x,z)}{d\pi_k} = \left(\sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\log(\pi_k) + \log(\text{Bern}(x_n|\theta_k))\cdot z_{nk}\right)'$
$= \lambda \sum_{n=1}^{N}\sum_{k=1}^{K} \frac{z_{nk}}{\pi_k} = 0$
$= \frac{N}{\pi_k} = 0$
$\Rightarrow \pi_k = 0$